

torch.signal.windows.blackman#

<https://pytorch.org/docs/stable/generated/torch.signal.windows.blackman.html>

Description:

Computes the Blackman window. The Blackman window is defined as follows: The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window. sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

Function Signature

```
torch.signal.windows.blackman(M, *, sym=True, dtype=None,
layout=torch.strided, device=None, requires_grad=False)
[source]#
```

Parameters

Keyword Arguments

Return type

Parameters

Keyword

`sym` (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. `dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`). `layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`. `device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. `requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric Blackman window.  
>>> torch.signal.windows.blackman(5)  
tensor([-1.4901e-08,  3.4000e-01,  1.0000e+00,  3.4000e-01,  
        -1.4901e-08])  
  
>>> # Generates a periodic Blackman window.  
>>> torch.signal.windows.blackman(5, sym=False)  
tensor([-1.4901e-08,  2.0077e-01,  8.4923e-01,  8.4923e-01,  
        2.0077e-01])
```

Detailed Documentation

Documentation

Computes the Blackman window.

The Blackman window is defined as follows:

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and `sym` is `True`.

`M` (int) – the length of the window. In other words, the number of points of the returned window.

`sym` (bool, optional) – If `False`, returns a periodic window suitable for use in spectral analysis. If `True`, returns a symmetric window suitable for use in filter design. Default: `True`.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if `None`, uses a global default (see `torch.set_default_dtype()`).

`layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

`device` (torch.device, optional) – the desired device of returned tensor. Default: if `None`, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: `False`.

